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CO2070 – COMPUTER ARCHITECTURE

Lab 4.5 - Building a Simple Processor Extended ISA

Overview

This report describes the changes made to the baseline processor to support six new instructions: mult, sll, srl, sra, ror, and bne. The modifications comply with the requirement to use the existing 3-bit ALUOP signal, ensure single-cycle execution within 8 time units, and design hardware logic without high-level behavioral operators.

Instruction Encodings & Assigned Opcodes

All new instructions use the standard 32-bit instruction word format. To avoid overlap with existing legacy operations and memory instructions like lwd/swd, the new opcodes were assigned in sequential binary order starting from 0x08.

Instruction	**Type**	**Bits 31:24 (Opcode)**	**Bits 23:16 (RD / Offset)**	**Bits 15:8 (RS1)**	**Bits 7:0 (RS2 / IMM)**
mult	R-Type	0000 1000 (0x08)	Destination Register	Source Register 1	Source Register 2
sll	I-Type	0000 1001 (0x09)	Destination Register	Source Register 1	Shift Amount (Imm)
srl	I-Type	0000 1010 (0x0A)	Destination Register	Source Register 1	Shift Amount (Imm)
sra	I-Type	0000 1011 (0x0B)	Destination Register	Source Register 1	Shift Amount (Imm)

ror	I-Type	0000 1100 (0x0C)	Destination Register	Source Register 1	Shift Amount (Imm)
bne	I-Type	0000 1101 (0x0D)	Branch Offset (Imm)	Source Register 1	Source Register 2

Control Unit Signal Matrix

The existing 3-bit ALUOP control signal supports a maximum of 8 functional units. With 000 to 011 already assigned to Forward, Add, And, and Or, the new math modules were assigned 100 (Multiplier) and 101 (Shared Shifter). To distinguish the four shift types, a simple SHIFT_SEL signal was introduced.

Opcode	ALUOP	WRITEENABLE	IMM_SEL	NEG_SEL	BRANCH_EQ	BRANCH_NEQ	SHIFT_SEL
mult	100	1	0	0	0	0	00
sll	101	1	1	0	0	0	00
srl	101	1	1	0	0	0	01
sra	101	1	1	0	0	0	10
ror	101	1	1	0	0	0	11
bne	001 (Sub)	0	0	1	0	1	00

Changes Made to Datapath & Control

To support the new instruction encodings without a major change to the datapath, the following specific changes were made:

PC Source Multiplexer Update (cpu.v):

The branch evaluation logic was expanded to support both equality (beq) and inequality (bne). The Program Counter update logic was revised to:

assign pc_src = jump | (branch_eq & zero) | (branch_neq & ~zero);

This effectively reuses the ALU's existing subtraction logic and ZERO flag.

Shift Selector (control_unit.v & alu.v):

A new 2-bit control signal, SHIFT_SEL[1:0], was routed from the Control Unit to the ALU. This allows the ALU to have only one shift unit, which meets the 8-unit hardware limit, while supporting four different shifting behaviors based on the selector value.

ALU Functional Unit Architecture

As per the exercise requirements, high-level Verilog behavioral operators (*, <<, >>) were strictly avoided.

The Multiplier Unit (mult_unit)

The multiplier calculates the product of two 8-bit inputs using an unrolled, combinational "shift-and-add" array.

- It generates 8 partial products using bit-selection and concatenation.
- These partial products are summed together via standard adders.
- It incurs a #2 latency, matching the standard addition unit.

The Universal Shifter (shift_unit)

To conserve functional unit slots inside the ALU, sll, srl, sra, and ror all share a single module.

- Logic: The module reads the 3 LSBs of DATA2 (the shift amount) and utilizes a 3-stage cascaded multiplexer network for each shift type.
- Mechanism: Rather than behavioral shifting, bit-concatenation is used (e.g., logical right shifting by 2 is implemented as {2'b00, DATA1[7:2]}).
- Multiplexing: The SHIFT_SEL signal dictates which of the four pre-calculated cascaded results is routed to the final output.
- It incurs a #1 latency.

Timing Analysis

The processor operates on an 8-time-unit clock cycle (1 clock cycle = 8 ns, given #4 toggles). It is critical that all new instructions resolve within this timeframe.

Multiplier Execution Path:

- Instruction Fetch: #2
- Control Unit Decode: #1
- Register File Read: #2
- ALU Multiply (Combinational): #2
- Register File Write (Synchronous setup): #1

Total Time: 8 time units (Safely completes within 1 clock cycle).

Shifter Execution Path:

- Instruction Fetch: #2
- Control Unit Decode: #1
- Register File Read: #2
- ALU Shift (Cascaded Muxes): #1
- Register File Write (Synchronous setup): #1

Total Time: 7 time units (Safely completes within 1 clock cycle).

Branch Execution Path (bne):

- Instruction Fetch: #2
- Control Unit Decode: #1
- Register File Read: #2
- ALU Subtraction (Flag Gen): #2
- Branch/Jump Target Adder (Parallel): #2 (Resolves concurrently with ALU)

Total Time: 7 time units (Leaves 1 time unit buffer before the next positive clock edge triggers the PC update).